

VPL: A Novel Cost Model for Earth and Mars-Orbit Navigation and Communication Constellations

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Abstract

We propose a cost estimating relationship (CER) for satellite constellations. The CER predicts the programme cost from platform mass M and unit count N . The dataset holds 25 systems from 1997 to 2025. Every model input carries a public source. The model writes the programme cost as development cost plus production cost on a learning curve, $\text{TOTAL} = g[\text{DEV}(M) + \text{PFM}(M) N^{\text{LRE}}]$. Leave-one-out cross-validation gives $R^2 = 0.945$. A second regime (New-Space) costs 5 to 21 times less at equal M and N . Unit-cost data does not identify the one-time development cost. Five programmes with disclosed budgets fix the development-to-production split by class and bound $\text{DEV}(M)$ to a factor of 1.2. On the traditional regime the CER slightly beats the SMAD USCM8+QuickCost benchmark ($R^2 = 0.88$ versus 0.87); for New-Space the benchmark over-predicts unit cost by a factor near 7. A planetary factor $F = 1.072$ carries the CER to Mars orbit.

Keywords: cost estimating relationship, satellite constellation, parametric cost model, learning curve

1. Introduction

Public unit-cost data covers few navigation and communication satellites. The cost models in SMAD use data up to the early 2000s [1]. This paper uses 25 systems launched from 1997 to 2025. The CER predicts programme cost from platform mass M and unit count N . Section 2 gives the data and its sources. Section 3 gives the earlier method, the new model, and the planetary factor. Section 4 gives the results and the comparison with the SMAD USCM8 and QuickCost models. Section 5 gives the limits.

2. Data

Table 1 lists the 25 systems and the source of each mass, unit count, and unit cost. AVG is the average unit cost. N is the unit count. M is the platform wet mass. All costs use FY2025 euros. USD converts to EUR at 0.9306 [2]. Earlier-year costs escalate to 2025 with US CPI [3]. For most systems the platform mass is 0.79 times the satellite mass [4]; where the manufacturer publishes the platform mass, the paper uses it directly. The traditional regime holds 20 systems. The New-Space regime holds 5 systems. The paper includes a system when a public per-unit cost exists.

Where a source gives a programme total, the paper derives the unit cost as the total divided by N and states the derivation. New-Space unit costs come from analyst estimates and are marked. Table 1 cites the mass, unit count and unit cost of every system separately.

Table 1: Dataset (single source of truth: the reproduction notebook). M is the platform wet mass. Reg.: T = traditional, NS = New-Space. * marks a New-Space analyst estimate. Mass, unit count and unit cost are cited separately.

System	M [kg]	N	AVG [€M]	Reg.
Thales HE-R1000	1000 [5]	5 [6]	106.0 [6]	T
GPS Block III	3065 [7]	10 [7]	268.6 [7]	T
Galileo 2 Generation	1817 [8]	12 [9]	133.5 [9]	T
GPS Block IIF	1290 [10]	12 [10]	179.1 [11]	T
GPS Block IIR	1604 [12]	13 [12]	64.2 [13]	T
Globalstar-3	395 [14]	17 [15]	21.9 [15]	T
O3b MEO	553 [16]	20 [17]	36.1 [18]	T
Globalstar 2nd Gen	553 [19]	24 [20]	48.0 [21]	T
Galileo FOC Overall	577 [22]	34 [23]	49.0 [24, 25]	T
GLONASS-K	739 [26]	5 [26]	110.7 [27]	T
GLONASS-M	1118 [28]	51 [28]	37.5 [27]	T
Iridium NEXT	679 [29]	81 [29]	36.9 [30]	T
Telesat Lightspeed	632 [31]	198 [32]	18.6 [33]	T
QZSS QZS-2/3/4	3239 [34]	7 [35]	218.6 [35]	T
ViaSat-3 F1	5070 [36]	3 [36]	664.4 [37]	T
Inmarsat-6	4321 [38]	2 [39]	385.1 [39]	T
AEHF	7110 [40]	6 [40]	965.0 [40]	T
WGS	4730 [41]	12 [42]	335.0 [41]	T
QZSS QZS-5/6/7	3792 [43]	3 [43]	255.6 [44]	T
O3b mPOWER	1343 [45]	8 [46]	158.0 [47]	T
OneWeb Gen-1	116 [48]	648 [49]	0.9 [50]*	NS
Amazon Kuiper	451 [51]	3232 [52]	1.9 [53]*	NS
Starlink v1.5	243 [54]	2987 [55]	0.4 [56]*	NS
Starlink v2-mini	577 [57]	7676 [55]	0.8 [58]*	NS

System	M [kg]	N	AVG [€M]	Reg.
AST BlueBird B2	4819 [59]	60 [60]	19.0 [61]*	NS

3. Method

3.1. Earlier method

The earlier method extracted a per-satellite production cost (PFM) from AVG, N , and M . It set the share $\lambda = \text{PFM}/(\text{PFM} + \text{DEV})$ as a linear function of N , from 0.35 at $N = 5$ to 0.15 at $N = 198$. It then fit $\text{PFM} = 0.149 M^{0.89}$. This method has two limits. First, the extraction removes N ; leave-one-out cross-validation of that CER gives $R^2 = 0.62$. Second, the function $\lambda(N)$ comes from a choice, not from data.

3.2. New model

The new model writes the programme cost as

$$\text{TOTAL}(M, N) = F g [\text{DEV}(M) + \text{PFM}(M) N^{\text{LRE}}], \quad (1)$$

with $\text{DEV}(M) = d M^p$, $\text{PFM}(M) = q M^r$, and $\text{LRE} = 1 + \ln(0.9)/\ln 2 = 0.848$. The learning rate is 0.9 [62]. The term g equals 1 for the traditional regime and 10^{off} for New-Space. The calibrated parameter *off* fits to -0.92 in log-space. F is the planetary factor (Section 3.3). The programme cost is development plus production: $\text{TOTAL} = \text{DEV} + \text{PROD}$, with $\text{PROD} = \text{PFM} N^{\text{LRE}}$. The average unit cost is $\text{AVG} = \text{TOTAL}/N$, so the identity $\text{TOTAL} = \text{AVG} \cdot N$ holds by construction.

Unit-cost data alone does not identify the one-time development cost DEV. A free fit of d and p gives a development band that runs to zero. The paper fixes this with disclosed budgets. Five programmes report development and production cost separately (Table 2). Their development fraction $\varphi = \text{DEV}/\text{TOTAL}$ is a ratio, so it constrains the split without leaving the dataset’s cost basis. The five parameters d, p, q, r, off are fit in one least-squares step against two sets of observations: the unit cost AVG of every system, and the disclosed φ of the four bespoke programmes. This keeps the cross-validated fit and bounds $\text{DEV}(M)$ (Section 4.2). The resolved CER is $\text{DEV}(M) = 5.99 M^{0.613}$ and $\text{PFM}(M) = 0.219 M^{0.828}$, with $g = 0.120$ for New-Space.

3.3. Planetary cost factor

The dataset holds Earth-orbit satellites. A Mars-orbit constellation costs more, so the paper adds a planetary factor F . The competition derived F bottom-up, at the subsystem level. For each spacecraft bus subsystem it takes a Mars cost-increase fraction from the literature and multiplies it by that subsystem’s share of the bus cost, then sums over the subsystems. The cost-increase fractions come from Mars-mission cost studies [63, 64]: the power subsystem needs about twice the solar-array area (Mars receives 45% of Earth’s solar flux) and larger batteries; the propulsion subsystem grows for Mars-orbit insertion; and the structure, thermal control and avionics carry radiation hardening and deep-space design margins. The subsystem cost shares come from the SMAD bus cost model. The weighted sum is a 7.2% increase in bus cost, with a 4.5% standard deviation across the subsystems. The paper adopts $F = 1.072$ for Mars and $F = 1$ for Earth. The SMAD QuickCost model applies a single 1.29 planetary factor [65]; the subsystem estimate is lower and traceable.

3.4. Validation

Cross-validation rotates the held-out data. Leave-one-out removes one system, refits, and predicts the removed system. Repeated 5-fold removes a set of systems. Bootstrap resampling gives coefficient intervals. The regime is a design input.

4. Results

4.1. Fit and cross-validation

Leave-one-out gives $R^2 = 0.945$ over the 25 systems and $R^2 = 0.845$ on the traditional regime alone. The 1-sigma prediction band is a factor of 1.55. The typical absolute error is 44%. Figure 1 shows predicted against actual cost.

4.2. Development split

For a 1 tonne system, the bootstrap gives $\text{DEV} = \text{€}414\text{M}$ with a 1-sigma factor of 1.2; the free fit was unbounded, running to zero. Development cost is 57% to 63% of total programme cost for the four bespoke programmes and 11% for the one commercial programme (WGS). Figure 2 shows the two CERs with their bootstrap bands and the disclosed data points each split in production and development.

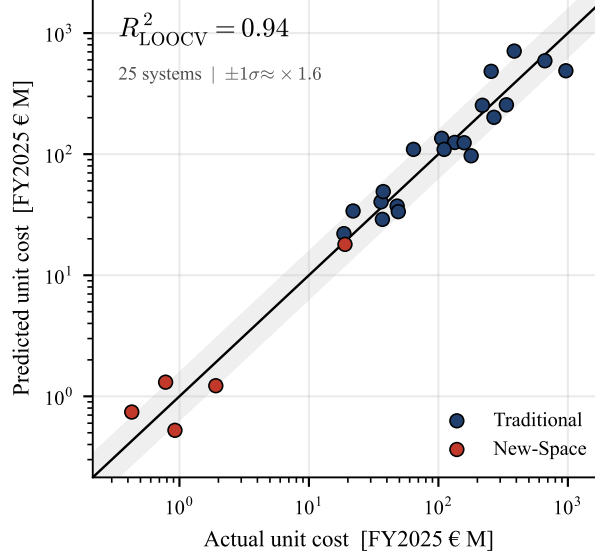


Figure 1: Predicted against actual unit cost. 25 systems. Leave-one-out $R^2 = 0.945$. Band: 1-sigma, factor 1.55.

Table 2: Disclosed development and production cost, as reported in the cited source (US programmes: then-year US\$M, SAR current estimate; Galileo: €M). $\varphi = \text{DEV}/(\text{DEV} + \text{PROD})$ is the development fraction and the only anchor quantity the fit uses; it is independent of currency and inflation basis. SAR: Selected Acquisition Report.

Programme	Development	Production	φ	Class	Source
AEHF	\$7465.9	\$5695.2	0.57	bespoke	[66]
MUOS	\$4140.1	\$2932.3	0.59	bespoke	[67]
SBIRS	\$10304.8	\$7176.9	0.59	bespoke	[68]
Galileo	€950.0	€566.0	0.63	bespoke	[69]
WGS	\$409.6	\$3392.3	0.11	commercial	[70]

4.3. New-Space regime

The New-Space regime costs 5 to 21 times less than the traditional regime at equal M and N . The point estimate is 8. The 5 New-Space systems carry masses from 116 to 4819 kg and unit counts from 60 to 7676.

4.4. Comparison with USCM8 and QuickCost

The paper compares the CER with the SMAD USCM8 development and production CERs and the QuickCost payload term [71, 72, 65]. On the traditional regime the CER is slightly more accurate: $R^2 = 0.88$ versus

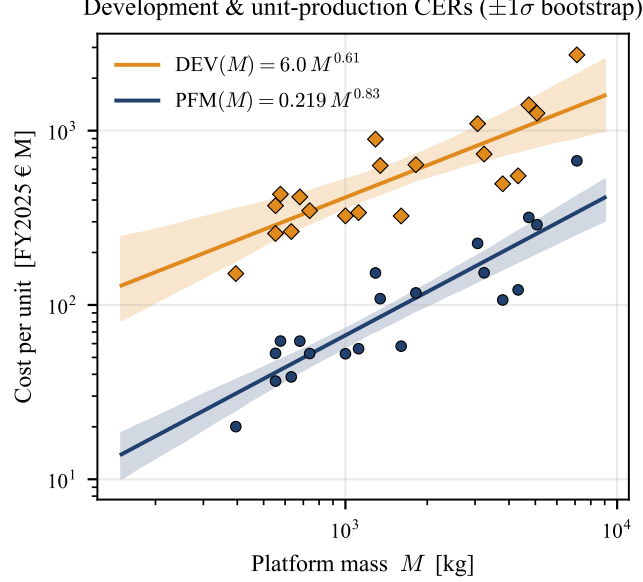


Figure 2: $DEV(M)$ and $PFM(M)$ CER components (lines) with 1-sigma bootstrap bands. Markers: each traditional system’s observed programme cost $AVG \cdot N$ split into development (diamonds) and unit production (circles) by the fitted class fraction. The development level is set by the disclosed fractions (Table 2); its band is a factor of 1.2.

0.87, and a median error of 24% versus 31%. The USCM8 CERs carry a standard error of 47% for development and 21% for production. On the New-Space regime USCM8+QuickCost over-predicts unit cost by a factor near 7, because it has no term for mass production. Figure 3 shows programme cost over M and N for the three streams. The traditional CER tracks the benchmark; the New-Space regime sits a full order of magnitude below it.

4.5. Operations

The paper carries the SMAD mission operations model for the operations phase [73]. For a 31-satellite constellation over 5 years it gives €159 M. The drivers are flight and ground software maintenance, 19 engineers and 4 technicians, and facilities.

5. Limitations

The dataset holds 25 systems. The New-Space regime holds 5 systems and is weakly cross-validated. New-Space unit costs come from analyst estimates. The development split depends on programme class. Four of the

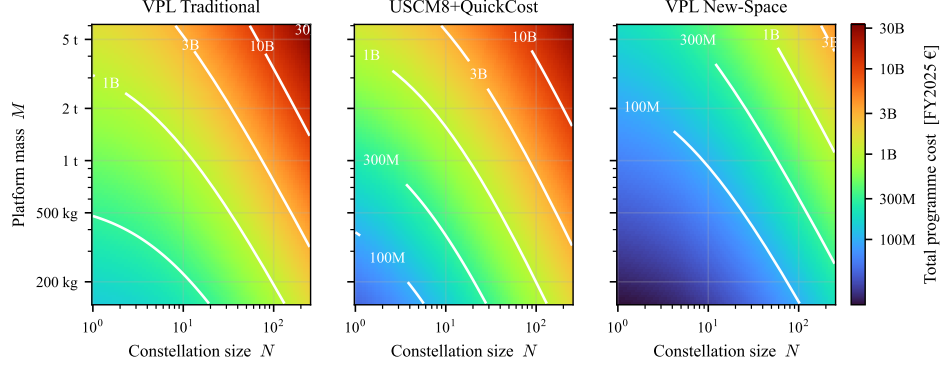


Figure 3: Total programme cost over platform mass and constellation size, with iso-cost contours, for the VPL Traditional CER, the SMAD USCM8+QuickCost benchmark, and the VPL New-Space regime.

five anchor programmes are bespoke; the commercial fraction rests on one anchor (WGS). Some descriptive attributes (power, spare count) have no public source and are omitted. The escalation to 2025 uses US CPI.

6. Conclusion

The CER predicts constellation programme cost from mass and unit count. Leave-one-out gives $R^2 = 0.945$. A second regime costs 5 to 21 times less. Disclosed budgets fix the development-to-production split and bound the development cost to a factor of 1.2. The CER matches the SMAD USCM8+QuickCost benchmark on traditional systems and corrects its over-prediction on New-Space systems.

References

- [1] Wertz, J. R. and Everett, D. F. and Puschell, J. J., Space Mission Engineering: The New SMAD, Microcosm Press, 2011.
- [2] Reference source, USD-EUR exchange rate history 2025, <https://www.exchange-rates.org/exchange-rate-history/usd-eur-2025>, accessed 2026-07-01 (n.d.).
- [3] Reference source, US Inflation Calculator, <https://www.usinflationcalculator.com/>, accessed 2026-07-01 (n.d.).

- [4] Reference source, Spacecraft platform/payload mass fraction - NASA NTRS, <https://ntrs.nasa.gov/api/citations/20140011472>, accessed 2026-07-01 (n.d.).
- [5] High Efficiency Radar Satellite HE-R 1000, Thales Group, <https://www.thalesgroup.com/en/solutions-catalogue/high-efficiency-radar-satellite-he-r-1000>, accessed 2026-07-02 (n.d.).
- [6] Korea Aerospace Industries 530 Million Dollar SAR Mission, <https://syntheticapertureradar.com/korea-aerospace-industries-530-million-dollar-sar-mission/>, accessed 2026-07-02 (n.d.).
- [7] Wikipedia, GPS Block III - Wikipedia, https://en.wikipedia.org/wiki/GPS_Block_III, encyclopedia; accessed 2026-07-01 (2026).
- [8] E. News, [Second-generation galileo satellite design has passed important review](https://www.engineeringnews.co.za/article/second-generation-galileo-satellite-design-has-passed-important-review) (Mar. 2022).
URL <https://www.engineeringnews.co.za/article/second-generation-galileo-satellite-design-has-passed-important-review-2022-03-09>
- [9] European Commission (DG DEFIS), Commission awards EUR 1.47 bn contracts to launch 2nd generation Galileo satellites, https://defence-industry-space.ec.europa.eu/commission-awards-eu147-bn-contracts-launch-2nd-generation-galileo-satellites-2021-01-20_en, government; accessed 2026-07-01 (2021).
- [10] Wikipedia, GPS Block IIF - Wikipedia, https://en.wikipedia.org/wiki/GPS_Block_IIF, encyclopedia; accessed 2026-07-01 (2024).
- [11] Behind the GPS IIF Launch, Inside GNSS, <https://insidegnss.com/behind-the-gps-iif-launch-a-long-and-winding-road/>, accessed 2026-07-02 (n.d.).
- [12] Wikipedia, GPS satellite blocks - Wikipedia, https://en.wikipedia.org/wiki/GPS_satellite_blocks, encyclopedia; accessed 2026-07-01 (n.d.).
- [13] Forecast International, Forecast International - GPS archive record, https://www.forecastinternational.com/archive/disp_pdf.cfm?DACH_RECNO=504, database; accessed 2026-07-01 (n.d.).

- [14] Via Satellite / SatelliteToday, MDA Wins 327M Satellite Order from Globalstar, Issues Bus Subcontract to Rocket Lab (Via Satellite), <https://www.satellitetoday.com/manufacturing/2022/02/24/mda-wins-327m-satellite-order-from-globalstar-issues-bus-subcontract-to-rocket-lab/>, news; accessed 2026-07-01 (2022).
- [15] MDA Space, MDA to Build 17 Satellites to Enhance Globalstar’s LEO Constellation, <https://mda.space/article/mda-to-build-17-satellites-to-enhance-globalstars>, manufacturer; accessed 2026-07-01 (2022).
- [16] SatNews, Arianespace’s Soyuz ST-B... The O3b Journey Has Begun (SatNews, quoting O3b/Thales), <http://satnews.com/2013/06/25/arianespaces-soyuz-st-b-o3b-its-a-launch-the-o3b-journey-has-begun-so-far-so-good-launch/>, news; accessed 2026-07-01 (2013).
- [17] Gunter Krebs / Gunter’s Space Page, O3b 13 - Gunter’s Space Page, https://space.skyrocket.de/doc_sdat/o3b-13.htm, encyclopedia; accessed 2026-07-01 (2019).
- [18] S. S.A., *Investor presentation (roadshow)*, may 2017 (May 2017).
URL https://www.ses.com/sites/default/files/2017-05/170510_ROAD%20SHOW%20PRESENTATION_FINAL.pdf
- [19] Globalstar-2 Launch Press Kit, Starsem, <https://www.starsem.com/news/images/Globalstar-2-press-kit.pdf>, accessed 2026-07-02 (n.d.).
- [20] Globalstar2 – Gunter’s Space Page, https://space.skyrocket.de/doc_sdat/globalstar-2.htm, accessed 2026-07-02 (n.d.).
- [21] Wikipedia, Globalstar - Wikipedia, <https://en.wikipedia.org/wiki/Globalstar>, encyclopedia; accessed 2026-07-01 (2025).
- [22] Galileo FOC (Full Operational Capability), <https://www.eoportal.org/satellite-missions/galileo-foc>, accessed 2026-07-02 (n.d.).
- [23] Wikipedia, *List of galileo satellites* (2026).
URL https://en.wikipedia.org/wiki/List_of_Galileo_satellites

- [24] B. Higgins, P. A. B. I. E. Subcommittee, [Galileo outline: Pnt advisory board international engagement subcommittee](#) (Dec. 2024).
URL https://www.gps.gov/sites/default/files/2025-06/AdvisoryMeetings_Higgins_Dec2024.pdf
- [25] Inside GNSS, [Ohb and sstl win second round of contracts for 8 galileo foc satellites](#), Inside GNSS (Feb. 2012).
URL <https://insidegnss.com/ohb-and-sstl-win-second-round-of-contracts-for-8-galileo-foc-satellites/>
- [26] Wikipedia, [Glonass-k](#) (2026).
URL <https://en.wikipedia.org/wiki/GLONASS-K>
- [27] The Jamestown Foundation, GLONASS Program for 2021-2030, <https://jamestown.org/program/glonass-program-for-2021-2030/>, other; accessed 2026-07-01 (2021).
- [28] GLONASS-M – Wikipedia, <https://en.wikipedia.org/wiki/GLONASS-M>, accessed 2026-07-02 (n.d.).
- [29] eoPortal (ESA), Iridium NEXT - eoPortal, <https://www.eoportal.org/satellite-missions/iridium-next>, space agency; accessed 2026-07-01 (n.d.).
- [30] Spaceflight Now, Iridium satellites rolling off assembly line in Arizona, <https://spaceflightnow.com/2016/07/13/iridium-satellites-rolling-off-assembly-line-in-arizona/>, news; accessed 2026-07-01 (2016).
- [31] Telesat Lightspeed – Wikipedia, https://en.wikipedia.org/wiki/Telesat_Lightspeed, accessed 2026-07-02 (n.d.).
- [32] Telesat, Telesat Contracts MDA as Prime Satellite Manufacturer, <https://www.telesat.com/press/press-releases/telesat-contracts-mda-as-prime-satellite-manufacturer-for-its-advanced-telesat-lightspeed-low-earth-orbit-constellation/>, press release; accessed 2026-07-01 (2023).
- [33] eoPortal (ESA), Telesat-Lightspeed - eoPortal, <https://www.eoportal.org/satellite-missions/telesat-lightspeed>, space agency; accessed 2026-07-01 (2024).
- [34] Qzs2 – Gunter’s Space Page, https://space.skyrocket.de/doc_sdat/qzs-2.htm, accessed 2026-07-02 (n.d.).

- [35] Qzss – Wikipedia, https://en.wikipedia.org/wiki/Quasi-Zenith_Satellite_System, accessed 2026-07-02 (n.d.).
- [36] Viasat3 – Gunter’s Space Page, https://space.skyrocket.de/doc_sdat/viasat-3.htm, accessed 2026-07-02 (n.d.).
- [37] ViaSat Suffers a 700 Million Dollar Setback, The Motley Fool, <https://www.fool.com/investing/2023/07/29/viasat-suffers-a-700-million-space-disaster/>, accessed 2026-07-02 (n.d.).
- [38] Inmarsat6 – Gunter’s Space Page, https://space.skyrocket.de/doc_sdat/inmarsat-6.htm, accessed 2026-07-02 (n.d.).
- [39] Reference source, Inmarsat-6 (I-6) Satellites - Airport Technology, <https://www.airport-technology.com/projects/inmarsat-6-i-6-satellites/>, expanded-dataset source; accessed 2026-07-01 (n.d.).
- [40] Aehf – Wikipedia, https://en.wikipedia.org/wiki/Advanced_Extremely_High_Frequency, accessed 2026-07-02 (n.d.).
- [41] Wgs11 – Wikipedia, <https://en.wikipedia.org/wiki/WGS-11>, accessed 2026-07-02 (n.d.).
- [42] Wgs – Wikipedia, https://en.wikipedia.org/wiki/Wideband_Global_SATCOM, accessed 2026-07-02 (n.d.).
- [43] Qzs6 – Wikipedia, <https://en.wikipedia.org/wiki/QZS-6>, accessed 2026-07-02 (n.d.).
- [44] R. Cheetham, [Jeo 5 – jmod goes shopping](#), Japan Earth Observer (May 2025).
URL <https://www.japanearthobserver.com/jeo-5-jmod-goes-shopping/>
- [45] O3Bmpower – Wikipedia, https://en.wikipedia.org/wiki/O3b_mPOWER, accessed 2026-07-02 (n.d.).
- [46] O3Bmpower – Gunter’s Space Page, https://space.skyrocket.de/doc_sdat/o3b-mpower1.htm, accessed 2026-07-02 (n.d.).
- [47] C. Henry, [Ses orders four more o3b mpower satellites from boeing](#), SpaceNews (Aug. 2020).
URL <https://spacenews.com/ses-orders-four-more-o3b-mpower-satellites-from-boeing/>

- [48] Oneweb – Gunter’s Space Page, https://space.skyrocket.de/doc_sdat/oneweb.htm, accessed 2026-07-02 (n.d.).
- [49] Oneweb – Wikipedia, https://en.wikipedia.org/wiki/OneWeb_satellite_constellation, accessed 2026-07-02 (n.d.).
- [50] How OneWeb plans to make its first satellites, SpaceNews, <https://spacenews.com/how-oneweb-plans-to-make-sure-its-first-satellites-arent-its-last/>, accessed 2026-07-02 (n.d.).
- [51] K. Gunter, Kuiper / amazon leo, Gunter’s Space Page (2026).
URL https://space.skyrocket.de/doc_sdat/kuipersat-1.htm
- [52] Orbital Radar, Amazon leo — leo broadband constellation (formerly project kuiper), Orbital Radar Library (May 2026).
URL <https://orbitalradar.com/satellites/operator/amazon-leo>
- [53] A. Boyle, Market study says amazon is spending up to \$20b on project kuiper satellite network, GeekWire (Sep. 2024).
URL <https://www.geekwire.com/2024/market-study-amazon-cost-project-kuiper-satellite-quilty/>
- [54] Starlink Group 6-53 (Falcon 9 Block 5), Everyday Astronaut, <https://everydayastronaut.com/starlink-group-6-53-falcon-9-block-5/>, accessed 2026-07-02 (n.d.).
- [55] Wikipedia, List of starlink and starshield launches (2026).
URL https://en.wikipedia.org/wiki/List_of_Starlink_and_Starshield_launches
- [56] E. Ralph, SpaceX announces second starlink satellite launch in two weeks, Teslarati (Dec. 2019).
URL <https://www.teslarati.com/spacex-starlink-satellite-launch-second-announcement/>
- [57] Starlink V2Mini – Gunter’s Space Page, https://space.skyrocket.de/doc_sdat/starlink-v2-mini.htm, accessed 2026-07-02 (n.d.).
- [58] Starlink soars: SpaceX satellite internet, SpaceNews, <https://spacenews.com/starlink-soars-spacexs-satellite-internet-surprises-analysts-with-6-6-billion-revenue-projection/>, accessed 2026-07-02 (n.d.).

- [59] G. D. Krebs, [Bluebird block 2](#), Gunter’s Space Page (2026).
URL https://space.skyrocket.de/doc_sdat/bluebird-2.htm
- [60] M. Donegan, [Ast spacemobile targets ‘intermittent’ national service in early 2026](#), Light Reading (Nov. 2025).
URL <https://www.lightreading.com/satellite/ast-spacemobile-targets-intermittent-national-coverage-in-early-2026>
- [61] AST SpaceMobile Secures Multi-Launch Deals, Spaceflight Now, <http://spaceflightnow.com/2024/11/17/ast-spacemobile-secures-multi-launch-agreements-with-blue-origin-isro-and-spacex/>, accessed 2026-07-02 (n.d.).
- [62] Reference source, Learning Rate Sensitivity Model - ICEAA, <https://www.iceaaonline.com/wp-content/uploads/2018/07/MM01-Learning-Rate-Sensitivity-Model-Anderson.pdf>, accessed 2026-07-01 (n.d.).
- [63] Fox, B. and Brancato, K. and Alkire, B., Guidelines and Metrics for Assessing Space System Cost Estimates, Tech. Rep. TR-418, RAND Corporation (2008).
- [64] Jones, H. W., Estimating the Life Cycle Cost of Space Systems, in: 45th International Conference on Environmental Systems, 2015.
- [65] Wertz, J. R., QuickCost model and planetary correction factor, Table 11-12, Space Mission Engineering: The New SMAD (2011).
- [66] US Department of Defense, AEHF Selected Acquisition Report (SAR), December 2013 – Total Acquisition Cost and Quantity, Total Program, https://www.globalsecurity.org/military/library/budget/fy2013/sar/01_aehf__december2013_sar.pdf, RDT and E 7465.9 / Procurement 5695.2 then-year USD M; see PDF p.17-18; accessed 2026-07-02 (2014).
- [67] US Department of Defense, MUOS Selected Acquisition Report (SAR), December 2012 – Total Acquisition Cost and Quantity, Total Program, https://www.globalsecurity.org/space/library/budget/fy2012/sar/muos_december_2012_sar.pdf, RDT and E 4140.1 / Procurement 2932.3 then-year USD M; see PDF p.15; accessed 2026-07-02 (n.d.).

- [68] US Department of Defense, SBIRS High Selected Acquisition Report (SAR), December 2011 – Total Acquisition Cost and Quantity, Total Program, https://www.globalsecurity.org/space/library/budget/fy2011/sar/sbirs-high_sar_31-dec-2011.pdf, RDT and E 10304.8 / Procurement 7176.9 then-year USD M (total, incl. Other Proc); see PDF p.12-17; accessed 2026-07-02 (n.d.).
- [69] European Space Agency and OHB System AG, Galileo in-orbit validation contract signed – ESA (IOV EUR 950M, 2006) and OHB FOC first order (EUR 566M, 2010), https://www.esa.int/About_Us/Business_with_ESA/Galileo_in-orbit_validation_contract_signed, IOV EUR 950M bundles development plus build of first 4 satellites and ground segment; FOC first order EUR 566M for 14 satellites; accessed 2026-07-02 (n.d.).
- [70] US Department of Defense, WGS (Wideband Global SATCOM) Selected Acquisition Report (SAR), December 2015 (DTIC AD1019569), <https://apps.dtic.mil/sti/tr/pdf/AD1019569.pdf>, RDT and E 409.6 / Procurement 3392.3 then-year USD M; see PDF p.13; accessed 2026-07-02 (n.d.).
- [71] Shao, A., USCM8 Nonrecurring CERs, Table 11-8 (companion workbook), Space Mission Engineering: The New SMAD; development SEE 47 percent (2011).
- [72] Shao, A., USCM8 Recurring T1 CERs, Table 11-9 (companion workbook), Space Mission Engineering: The New SMAD; production SEE 21 percent (2011).
- [73] Shao, A., Mission Operations Cost Prediction Model, Table 11-28, Space Mission Engineering: The New SMAD (2014).